ModelIt!

Student Activity Pack

Why Is Groundwater Disappearing?

How Human Activity and Natural Cycles Determine the Future of Underground Water

Grade 08L2 | NGSS MS-ESS3-1, MS-ESS2-4 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the water come from when you turn on a tap? Trace it as far back as you can.

2 What is an aquifer, and how does water get underground?

3 Why might pumping too much water from underground be a problem?

4 Draw a simple diagram showing how rain becomes groundwater and how wells pump it back up.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Agricultural Irrigation | Rainfall Recharge | Aquifer Level | Land Subsidence | Long-term Water Security

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Balanced vs. Over-Extraction

Equal pumping and recharge vs. heavy pumping with low recharge

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought Impact

Moderate irrigation with very low rainfall recharge

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Long-term Projection

Heavy irrigation sustained for 50 years

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Urban Water Demand — Water consumed by cities and towns for drinking, sanitation, and industry — competing with agricultural use

Aquifer Recharge Rate — The speed at which water naturally percolates down to the aquifer — varies by soil type and geology

Contamination Risk — The chance that pollutants enter the aquifer as water levels drop and flow patterns change

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are aquifers that took thousands of years to fill being drained in just decades? Use evidence from your simulation to support your answers.

1. **Why is groundwater depletion considered a 'slow-motion crisis' that many people don't notice until it's too late?**

2. **How does land subsidence create a feedback loop that makes the problem worse?**

3. **What responsibilities do current generations have to preserve groundwater for future generations?**

4. **How does climate change affect the balance between rainfall recharge and irrigation demand?**

5. **Why is it harder to manage groundwater than surface water like rivers and lakes?**

8 STEM Challenge: Design a Sustainable Water Management Plan

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YOUR ENGINEERING MISSION

Design an evidence-based water management plan for a farming community that balances agricultural needs with long-term aquifer sustainability, using your model data to justify every recommendation.

THE SCENARIO

A farming community in the Central Valley of California depends on groundwater for 80% of its irrigation. The aquifer level has dropped 60 feet in the last 30 years, and land subsidence has damaged roads and canals. Your team must create a plan that sustains both farming and the aquifer.

Engineering Design Thinking

1. WHAT SPECIFIC IRRIGATION REDUCTION TARGETS DOES YOUR MODEL EVIDENCE SUPPORT?
2. HOW WILL YOU BALANCE SHORT-TERM FARMING NEEDS WITH LONG-TERM WATER SECURITY?
3. WHAT INFRASTRUCTURE CHANGES COULD INCREASE RAINFALL RECHARGE RATES?
4. HOW WILL YOU MONITOR AQUIFER LEVELS AND LAND SUBSIDENCE TO MEASURE YOUR PLAN'S EFFECTIVENESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrogeologists and Water Resource Engineers study groundwater systems and design sustainable water management strategies for communities, agriculture, and industry. They earn $65,000-$115,000/year.